

ECO102: Introduction to Macroeconomics

Lecture 7: Aggregate Supply and Aggregate Demand *(Chapter 27: Aggregate Supply and Aggregate Demand)*

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Chapter 27: Aggregate Supply and Aggregate Demand, pp 647 – 653

What is the AS-AD Model

- We have seen the relationship between the labour input and output (Real GDP) in our previous chapters
- Now, in our pursuit of looking at the economy in a different light, we will look at the ***Aggregate Supply-Aggregate Demand Model (AS-AD)***
- The AS-AD Model charts the relationship between the ***output (Real GDP)*** and the ***price level***
- This model represents the consensus view of macroeconomists on how real GDP and the price level are determined
- The model provides a framework for understanding the forces that make our economy expand, that bring inflation, and that cause business cycle fluctuations

What is the AS-AD Model

- The price level and the output level in the AS-AD model are determined similarly to how the quantity and price in a competitive market are determined
- However, there are some differences, as the AS-AD model is a model of an imaginary market for the total of all the final goods and services that make up real GDP
- Despite it being an imaginary market, the AS-AD model makes clear distinctions between **supply/demand** and **quantity supplied/demanded**

Aggregate Supply

- The ***quantity of real GDP supplied*** is the total quantity of goods and services, valued in constant base year dollars, that firms plan to produce during a given period
- This quantity depends on the quantity of labour employed, the quantity of physical and human capital, and the state of technology
- ***Aggregate supply (AS)*** is the relationship between the quantity of real GDP supplied and the price level

Aggregate Supply

- At any given time, *the quantity of capital, the state of technology, and the population are fixed; only the quantity of labour changes*
- The labour market can be either at full employment, above full employment, or below full employment
- At full employment, the quantity of real GDP supplied is potential GDP
- Over the business cycle, employment fluctuates around full employment, and the quantity of real GDP supplied fluctuates around potential GDP
- We usually distinguish AS between two time frames:
 - **Short-run aggregate supply**
 - **Long-run aggregate supply**

Long-Run and Short-Run Aggregate Supply

- ***Short-run aggregate supply*** is the relationship between the quantity of real GDP supplied and the price level when the money wage rate, the prices of other resources, and potential GDP remain constant
- The **Short-run aggregate supply** is determined similarly to how we have determined our supply curve for a competitive market

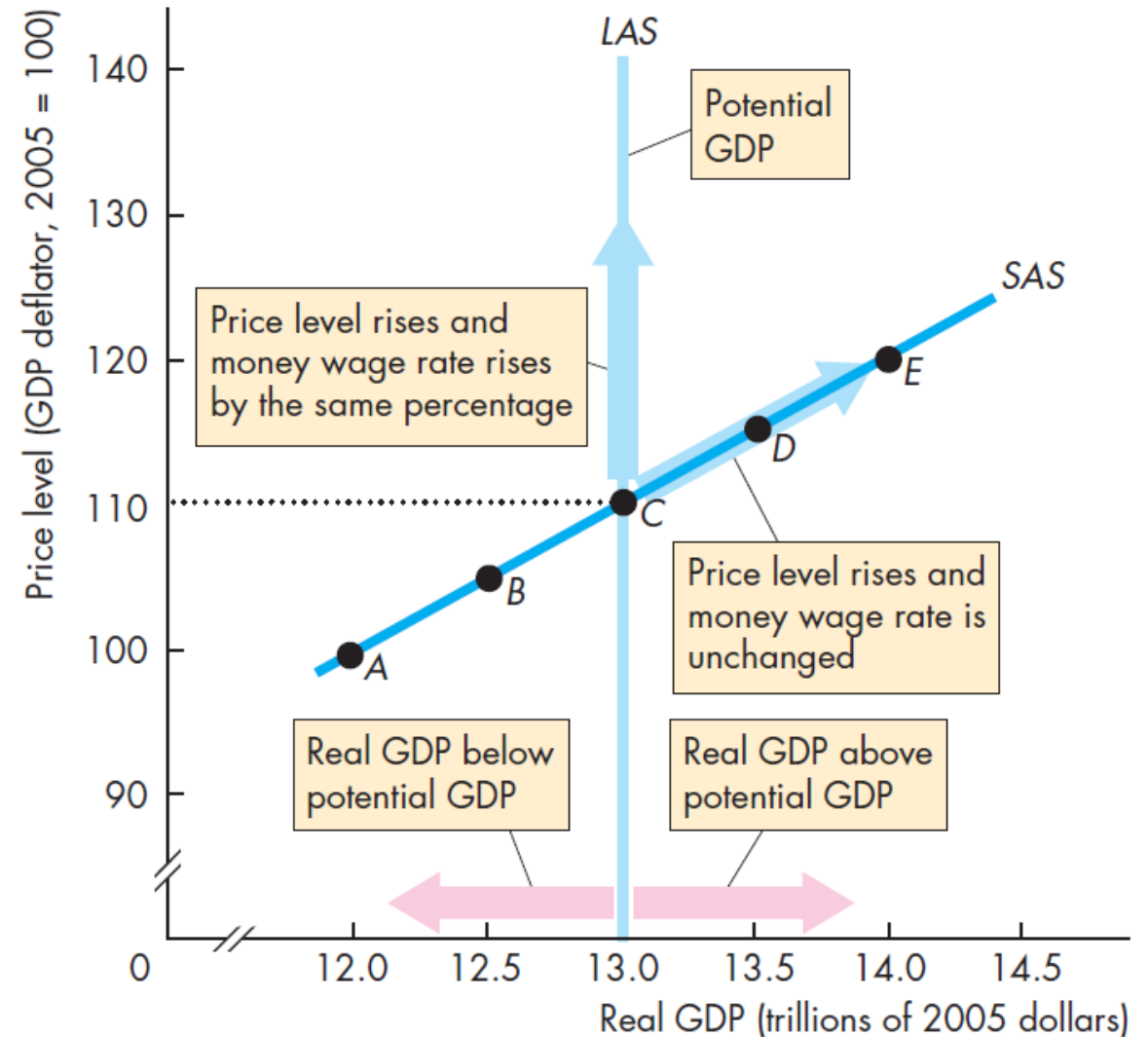
Long-Run and Short-Run Aggregate Supply

- ***Long-run aggregate supply*** is the relationship between the quantity of real GDP supplied and the price level when the money wage rate changes in step with the price level to maintain full employment
- The long-run aggregate supply curve is always vertical and is always located at potential GDP
- It is vertical because potential GDP is independent of the price level
- A 10 percent increase in the prices of goods and services is matched by a 10 percent increase in the money wage rate, causing the real wage rate to remain unchanged at its full-employment equilibrium level

Long-Run and Short-Run Aggregate Supply

In the long run, the quantity of real GDP supplied is potential GDP and the LAS curve is vertical at potential GDP

In the short-run, the quantity of real GDP supplied increases if the price level rises, while all other influences on supply plans remain the same



Changes in Aggregate Supply

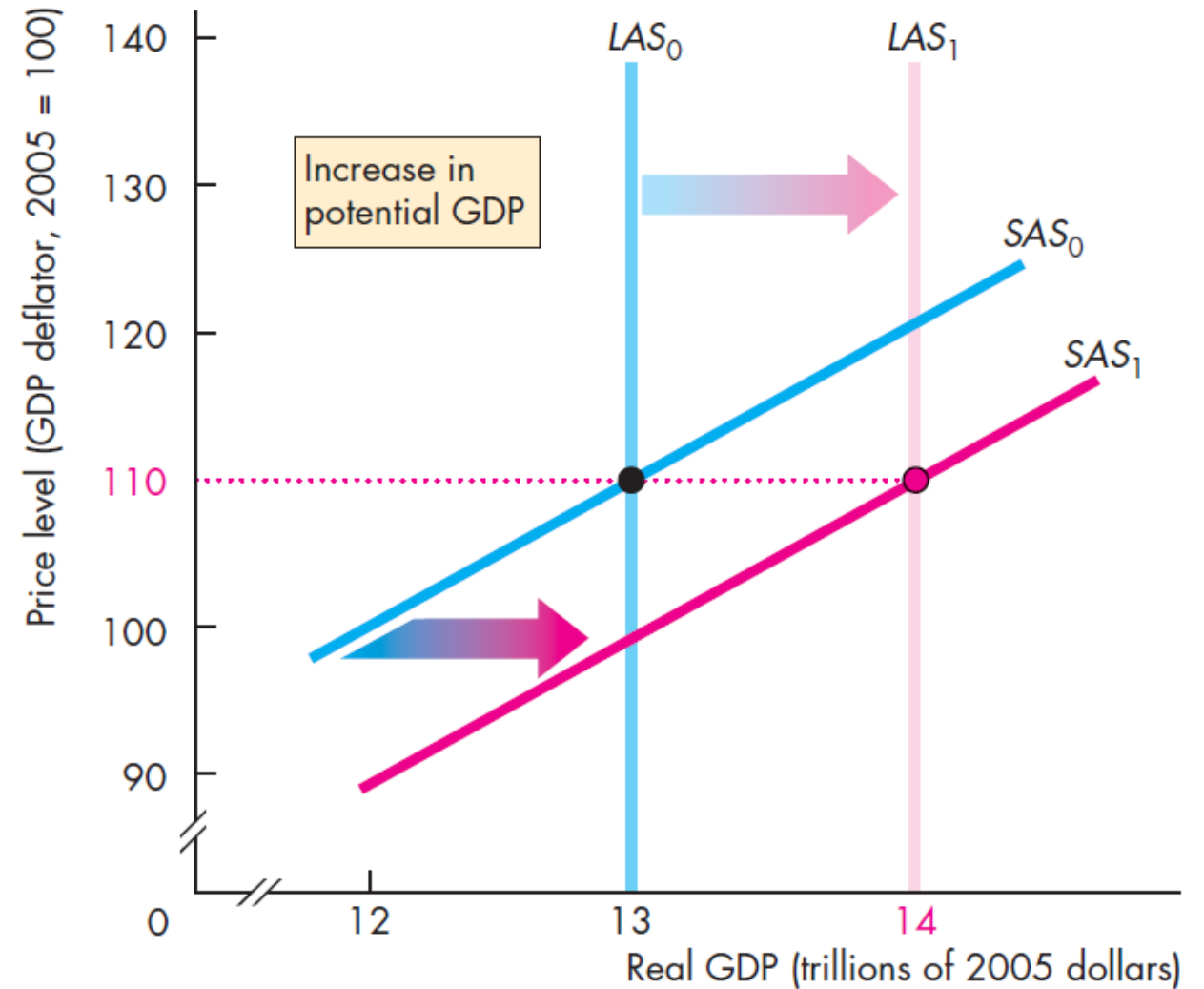
- **Aggregate supply** changes when an influence on production plans other than the price level changes
- These other influences include:
 - **Changes in potential GDP:** An increase in potential GDP increases both long-run aggregate supply and short-run aggregate supply
 - **Changes in the money wage rate:** When the money wage rate changes, short-run aggregate supply changes, but long-run aggregate supply does not change

Aggregate Supply: Changes in Potential GDP

- Potential GDP can increase for any of three reasons:
 - **An Increase in the Full-Employment Quantity of Labour:** Over time, potential GDP increases because the labour force increases; however, with constant capital and technology, potential GDP increases only if the full-employment quantity of labour increases
 - **An Increase in the Quantity of Capital:** The larger the quantity of capital, the more productive is the labour force and the greater is its potential GDP.
 - **An Advancement in Technology:** Technological advances are by far the most important source of increased production over the past two centuries

Aggregate Supply: Changes in Potential GDP

The long-run aggregate supply curve shifts rightward from LAS_0 to LAS_1 and the short-run aggregate supply curve shifts from SAS_0 to SAS_1

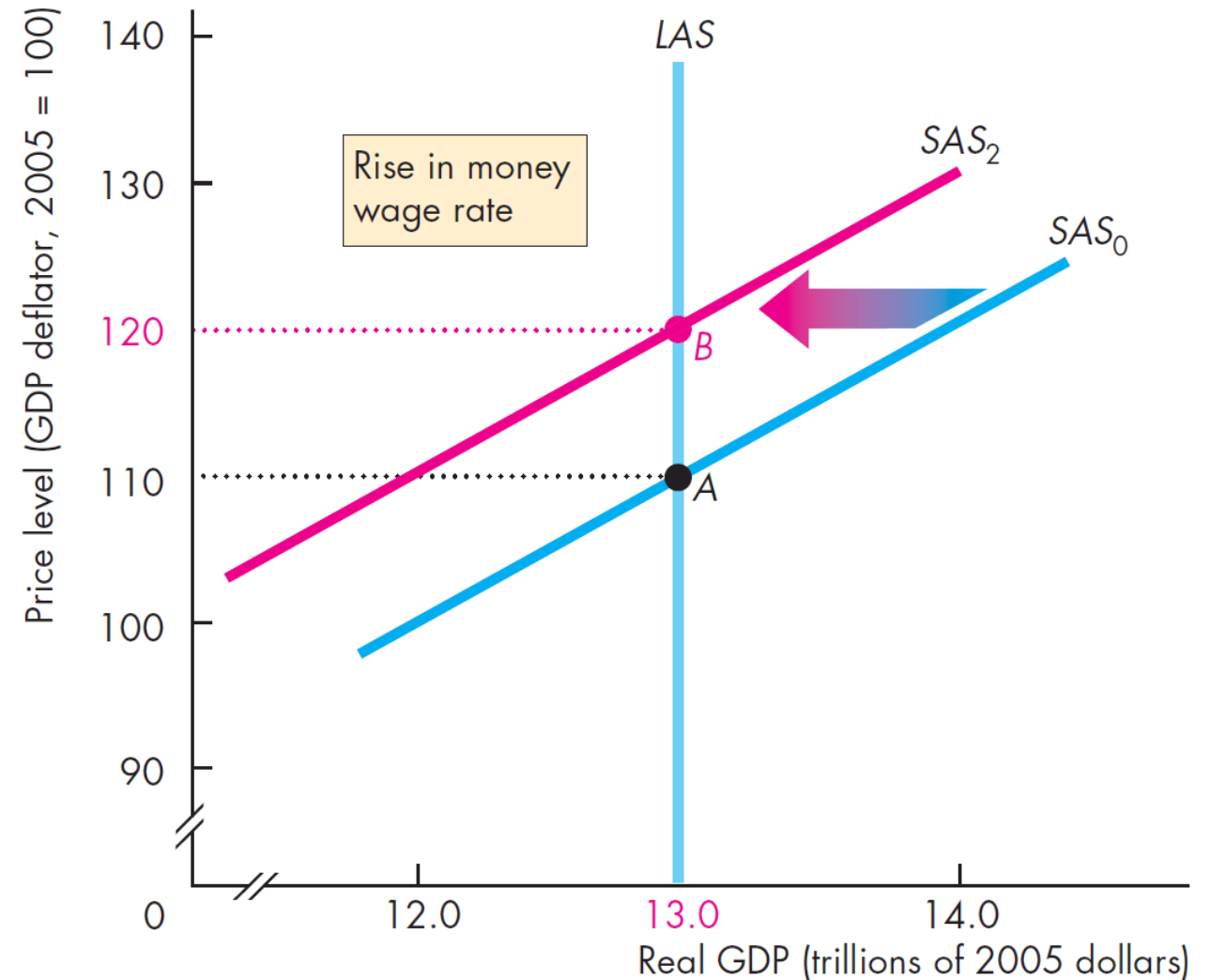


Aggregate Supply: Changes in the Money Wage Rate

- The money wage rate can change for two reasons:
 - **Departures from full employment:** Unemployment above the natural rate puts downward pressure on the money wage rate, and unemployment below the natural rate puts upward pressure on it
 - **Expectations about inflation:** An expected rise in the inflation rate makes the money wage rate rise faster, and an expected fall in the inflation rate slows the rate at which the money wage rate rises

Aggregate Supply: Changes in the Money Wage Rate

A rise in the money wage rate decreases short-run aggregate supply and shifts the short-run aggregate supply curve leftward from SAS_0 to SAS_2



Aggregate Demand

- The ***quantity of real GDP demanded*** is the total amount of final goods and services produced in a Country that people, businesses, governments, and foreigners plan to buy
- So, the ***quantity of real GDP demanded is determined by:***

$$Y = C + I + G + X - M$$

- Buying plans depend on many factors, including
 - The Price level
 - Expectations
 - Fiscal policy and monetary policy
 - The world economy

Aggregate Demand

- Other things remaining the same, the higher the price level, the smaller is the ***quantity of real GDP demanded***; and this relationship between the quantity of real GDP demanded and the price level is called aggregate demand
- ***Aggregate Demand*** is a downward-sloping curve similar to the case of the competitive market
- The aggregate demand curve slopes downward for two reasons:
 - ***Wealth effect***
 - ***Substitution effects***

Aggregate Demand

- **Wealth effect:** When the price level rises but other things remain the same, real wealth decreases. In order to restore their wealth, people increase their savings and, equivalently, decrease current consumption, leading to a decrease in aggregate demand
- **Substitution effects:** When the price level rises and other things remain the same, interest rates rise. Faced with a higher interest rate, people and businesses delay plans to buy new capital and consumer durable goods and cut back on spending.

This substitution effect involves changing the timing of purchases of capital and consumer durable goods and is called an intertemporal substitution effect

A second type of substitution effect occurs when, due to the increase in relative prices, people choose foreign-made products as opposed to locally produced products.

Aggregate Demand

A change in the price level, when all other influences on aggregate buying plans remain the same, brings a change in the quantity of real GDP demanded and a movement along the AD curve

